Supplemental Material: What a radical-pair quantum reservoir would require

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Every number below is computed under the single protocol used in the main text — correlated-pair birth, converged time grid, out-of-sample capacity estimation — with two exceptions that are labelled where they appear and retained deliberately as controls: the first row of Table S1, which is the superseded single-electron reset, and Table S7, which is the originally submitted single-realisation clock scan. The sections cover (S1) validation of the Liouville-space solver, (S2) the capacity estimator and its controls, (S3) the injection map and why a single-electron reset fails, (S4) the chemically accessible readout routes, their route-specific memoryless floors, their grid convergence and a re-run of the adverse analyses with the product register, (S5) the coherence fraction at matched hopping rate, (S6) classical baselines, (S7) the turnover clock and the register-reuse criterion, (S8) a semiclassical reference model for the coherence fraction, (S9) which nuclei can actually hold the register, (S10) ensemble averaging over heterogeneous and desynchronised copies, (S11) anisotropic hyperfine coupling and the nuclear spin quantum number, (S12) prediction of the two constants the criterion depends on, and (S13) the spin parameters. Sections S8–S12 were added in response to referee comment and, unlike the rest, several of them *reduce* the claims of the main text; we have kept the reductions rather than the original numbers. All are reproducible from the open `qbscreen` package.

S1. LIOUVILLE-SPACE SOLVER AND THE COHERENT LIMIT

The propagator is $e^{\mathcal{L}\tau}$ of the Haberkorn master equation assembled in Liouville space with the column-stacking convention $\mathrm{vec}(A\rho B) = (B^\top \otimes A)\,\mathrm{vec}(\rho)$; the time-integrated singlet yield follows from a single linear solve,

$$\Phi_S = k_S\,\mathrm{vec}(P_S)^\dagger\,(-\mathcal{L})^{-1}\,\mathrm{vec}(\rho_0). \tag{S1}$$

Adding the decoherence term must not corrupt the coherent dynamics. For $1/T_2^e \to 0$ with $k_S = k_T = k$, Eq. (S1) must reproduce the closed-form eigenbasis result $\Phi_S^{\rm coh} = k\big[\sum_m P_{mm}\rho_{mm}/k + \mathrm{Re}\sum_{m\neq n} P_{mn}\rho_{nm}/(k+i\omega_{mn})\big]$. For a two-electron-plus-nucleus model at $B = 0.9$ mT, $A = 200$ MHz and $k = 1\ \mu\mathrm{s}^{-1}$ the two agree to better than 10^{-6}; this is asserted as a unit test.

S2. CAPACITY ESTIMATOR AND ITS CONTROLS

For a target y_k of the input history the capacity is the out-of-sample squared multiple correlation: a ridge readout ($\lambda = 10^{-6}$) is fitted on the first half of the run and scored on the held-out second half. The totals are $\mathrm{MC} = \sum_{d\geq 0} C[s_{k-d}]$ and $\mathrm{IPC} = \mathrm{MC} + \sum_{\rm nonlin} C[y]$ over the degree-2 Legendre family (diagonal targets to $d_{\max} = 14$, cross targets for $d_1 < d_2 \leq 6$).

Null floor. Scoring the same feature matrices against targets built from an *independent* input sequence, for which the true capacity is zero, returns $+0.010$ for the estimator used here. The same sum without the $\max(0,\cdot)$ clip returns -1.49, i.e. it is far more biased, in the opposite direction. We therefore retain the clip and quote the floor rather than removing it.

Convergence. The capacity is converged in the length L of the driving sequence (six seeds, engineered point): $\mathrm{IPC} = 5.28 \pm 0.26$, 5.60 ± 0.20, 5.87 ± 0.19 and 5.84 ± 0.10 for $L = 600$, 1200, 2400 and 4800, i.e. flat beyond $L \approx 2400$. Under the superseded single-electron-reset protocol the same estimator did *not* converge, rising from 5.9 to 8.1 over the same range; convergence is a property of the corrected drive, not of the estimator.

The Dambre bound is not an independence test. $\mathrm{IPC} \leq n_{\rm obs}$ holds asymptotically for an orthonormal target family but is violated in finite samples: for $X = s$ ($n_{\rm obs} = 1$, $T = 1200$) the estimator returns 1.013 ± 0.009 on average and exceeds the bound in 29 of 30 seeds. (The submitted version quoted 1.011 and 30 of 30 from a run whose output was not retained; the figures here are from the stored, regenerable control `s2_dambre_control.json`, and we quote those rather than the unverifiable originals.) Satisfying the inequality therefore does not demonstrate that the readout observables are independent, and we make no such argument anywhere in this work.

S3. THE INJECTION MAP

A protocol that resets only the input electron to a product state while retaining the reduced state of the rest is not a chemical process. For a separated pair ($J = 0$) it destroys the electron–electron correlation on every cycle, and since $\langle P_S\rangle = \frac{1}{4} - \langle \boldsymbol{S}_1\cdot\boldsymbol{S}_2\rangle$ with $\langle \boldsymbol{S}_2\rangle = 0$, the singlet probability is pinned at exactly $1/4$ independently of the input. We replace it by correlated-pair birth: the nuclei are retained and the electron pair is recreated as $\rho_e(s) = s\,|S\rangle\langle S| + (1-s)P_T/3$, or as a coherent S–T_0 superposition. Table S1 gives the consequence at the cryptochrome point ($B = 50$ μT, $J = 0$, $\tau = 1$ μs, six seeds, kinetic-only readout).

TABLE S1. The injection map and the capacity it permits.

injection	max. flux	IPC
single-electron reset	2.7×10^{-14}	0.000 ± 0.000
correlated-pair birth, S/T	4.9×10^{-1}	2.941 ± 0.108
correlated-pair birth, S–T_0	4.3×10^{-1}	3.116 ± 0.209

Under the superseded map the spread of the observable is at the level of machine precision: the readout is a constant, and the vanishing capacity is an identity of the map rather than a statement about spin chemistry.

S4. CHEMICALLY ACCESSIBLE READOUT ROUTES

Recombination is switched on ($k_S = 1$, $k_T = 0.2$ μs^{-1}) so that the observable is the yield actually formed, $Y_X = k_X \int_0^\tau \mathrm{Tr}[P_X \rho(t)]\,\mathrm{d}t$, sampled at five laboratory times. CIDNP is the product-weighted nuclear polarisation, $\langle I_z\rangle_{\mathrm{prod}} = \int k_S \mathrm{Tr}[I_z P_S \rho]\,\mathrm{d}t \,/ \int k_S \mathrm{Tr}[P_S \rho]\,\mathrm{d}t$: the polarisation carried away by the recombined molecule, not that of the surviving pair.

TABLE S2. Capacity recovered by chemically accessible observables (six seeds).

readout route	channels	IPC
end-point singlet yield	1	1.017
both exit channels, end point	2	1.374
three micro-environments (different τ)	3	2.557
time-resolved kinetics, one product pool	5	2.008
both exit channels, time-resolved	10	2.012
kinetics + accumulated downstream pool	10	2.706
kinetics + CIDNP on the product	8	4.650

All seven routes defined by the model are listed; the submitted version omitted the fourth, and the last two rows are separate constructions rather than the cumulative additions the earlier indentation implied (the CIDNP route is five kinetic channels plus three polarisation channels, hence eight, not eighteen). The omitted route is informative: doubling the time-resolved channel count from five to ten by adding the second exit channel changes the capacity by 0.003, so the readout saturates in channel number.

Route-specific memoryless floors. The floor is not common to these routes and is measured by wiping the register every cycle ($q = 1$), which leaves only what each route computes from the present input (six seeds): 1.019 ± 0.025 for the one-, two-, three-, five- and ten-channel yield routes (including the heterogeneous-lifetime route, whose floor is measured by the same $q = 1$ construction), 2.005 ± 0.017 for the eight-channel CIDNP route, and 1.960 ± 0.070 for the accumulated pool. The CIDNP excess is therefore $4.650 - 2.005 = 2.64$, not the factor of 4.6 that a universal floor of unity would imply. The accumulated-pool route retains MC = 1.937 at $q = 1$ because a leaky integrator is itself a classical delay line; its capacity is not evidence of nuclear-spin memory and we exclude it from the claims in the main text.

The register is the product. The cycle above hands the next turnover the renormalised state of the pair that did *not* recombine. Rebuilding it so that the register is the nuclear state carried away by the recombined molecule, $\sigma_{\mathrm{nuc}} \propto \int [k_S \mathrm{Tr}_e(P_S \rho P_S) + k_T \mathrm{Tr}_e(P_T \rho P_T)]dt$, raises every capacity: kinetics $2.008 \to 2.742$, both-exit time-resolved $2.012 \to 2.757$, CIDNP $4.650 \to 4.909$, accumulated pool $2.706 \to 3.872$ (six seeds). The reported numbers are thus conservative with respect to this choice.

Do the adverse results survive the product register? They do. The sensitivity analyses of Secs. S9–S11 are quoted for the survivor register, so that they are directly comparable with the published pipeline; each has also been re-run with the product register.

TABLE S3. The reductions are not artifacts of the survivor register.

quantity	survivor	product
proton register, excess retained	29%	30%
proton register, kinetic excess	+0.000	+0.000
kinetic excess at 2 cycles of jitter	+0.004	−0.006
CIDNP excess at 2 cycles of jitter	+1.310	+1.153
anisotropy $\eta = 0.5$, fraction retained	32%	40%
anisotropy $\eta = 1$ (Ising), excess	0.000	0.005

Every reduction survives, and none of the paper's conclusions depends on the choice. The one visible difference is that the kinetic readout degrades more gracefully under moderate jitter with the product register (excess +0.317 at half a cycle, against +0.006), though it still reaches zero by two cycles. Anisotropy is likewise slightly less punishing (40% retained rather than 32%) while remaining a two-thirds loss, and the Ising point still gives an excess of 0.005, i.e. zero within the estimator's floor.

Time-grid convergence. A 40 MHz hyperfine coupling has a 25 ns period, so a coarse grid under-samples the dynamics and inflates the time-resolved capacities. Table S4 shows convergence from $n_t = 48$ onward; the main text uses $n_t = 96$ (dt $\approx$ 10 ns).

TABLE S4. Convergence in the number of time steps per cycle (six seeds, $L = 700$, the same protocol as Table S2).

n_t	dt (ns)	kinetics IPC	+CIDNP IPC
24	41.7	2.756	4.879
48	20.8	2.050	4.614
96	10.4	2.008	4.650
192	5.2	1.988	4.656

Why CIDNP is not an optional extra channel. The reduced state of either electron of a newly born pair is maximally mixed for every input, because both $|S\rangle\langle S|$ and $P_T/3$ carry zero one-electron polarisation; the input resides entirely in the two-electron correlation. With $J = 0$ the radicals do not interact, so nothing transfers that correlation to a nucleus. Consistently, with recombination switched off the cryptochrome-point capacity is 1.01, the memoryless floor. Spin-selective recombination is what converts the singlet–triplet correlation into a population difference and so writes the input into the nuclear register: in a separated pair it is the write mechanism, not merely the read mechanism.

S5. COHERENCE FRACTION AT MATCHED HOPPING RATE

An all-spin pure-dephasing rate γ is added in the S_z product basis. Taking $\gamma \to \infty$ at fixed τ measures Zeno freezing rather than a classical limit, because the golden-rule transfer rate scales as $|H_{ij}|^2/\gamma$; we therefore hold rate $\times\, \tau$ fixed by scaling $\tau \propto \gamma$, whereupon the capacity saturates.

TABLE S5. Incoherent limit at matched hopping rate (six seeds).

γ (MHz)	τ (ns)	IPC
0 (coherent)	50	5.421
10^3	10^3	0.999
5×10^3	5×10^3	0.998
2×10^4	2×10^4	0.998

The saturation value 1.00 is the memoryless floor, so the coherence fraction is $(5.421 - 0.999)/5.421 = 0.82$.

The added γ above acts on all five spins, which raises the objection that the collapse merely records erasure of the nuclear register rather than loss of electronic coherence. It does not. Restricting the added dephasing to the two electrons, leaving the nuclei entirely untouched, and holding the same matched τ gives IPC $= 0.9949 \pm 0.0222$, 0.9891 ± 0.0223 and 0.9891 ± 0.0224 at $\gamma = 10^3$, 5×10^3 and 2×10^4 MHz (six seeds), i.e. a coherence fraction of 0.817 against 0.816 for the all-spin channel. The quantity is a statement about the electrons.

It remains, however, an *operational* fraction: it is defined by one particular incoherent limit in which τ is stretched by up to a factor of 400, not by comparison against a physically motivated classical model. The appropriate reference would be a semiclassical radical-pair treatment with classical hyperfine vectors and rate equations [Timmel *et al.*, Mol. Phys. **95**, 71 (1998)]. Section S8 supplies exactly that reference and finds the classical model at its own memoryless floor, so 0.82 is conservative; we nonetheless do not present it as a measure of "how quantum" the reservoir is.

S6. CLASSICAL BASELINES

Tanh echo-state networks (spectral radius 0.9, input scale 0.6) are driven by the same input and scored with the same estimator (eight seeds, $L = 1200$).

TABLE S6. The quantum reservoir against matched classical reservoirs.

reservoir (readout features)	IPC
quantum, 5 spins (12 observables)	5.593 ± 0.191
ESN, 5 nodes (5 linear)	4.552 ± 0.325
ESN, 5 nodes (12: 5 linear + 7 quadratic)	9.414 ± 0.860
ESN, 12 nodes (12 linear)	10.366 ± 0.590

Matched by physical unit the quantum system is ahead; matched by readout dimension it is behind. We report both and claim no advantage. Which accounting is fair depends on a cost model for non-commuting measurements against multiplications that we do not have.

S7. TURNOVER CLOCK AND THE REGISTER-REUSE CRITERION

The memory horizon is (MC$-C_0$) times the cycle time, the zero-delay term $C_0 \simeq 1$ contributing no memory; the raw MC $\times\, T$ used in Table S7 below is the superseded convention and is retained only because that table is the originally submitted scan. Because the nuclei leave in the diamagnetic product the cycle time is the turnover interval T_d and not the radical-pair lifetime. We insert a pause between turnovers and relax the register with probability $q = 1 - \exp(-T_\mathrm{d}/T_{1n})$, taking $T_{1n} = 1$ s. That value is retained here for continuity with the submitted version; Sec. S12 shows it is a high-field number and that the correct low-field value for a protein-bound proton is some four hundred times shorter, which is what ultimately binds the criterion.

TABLE S7. The horizon follows the turnover interval.

T_d	q	MC (5 ch)	MC (8 ch)	horizon (8 ch)
1 μs	10^{-6}	1.948	2.818	5.6 μs
1 ms	10^{-3}	1.948	2.818	2.8 ms
10 ms	10^{-2}	1.947	2.817	28 ms
100 ms	9.5×10^{-2}	1.944	2.800	280 ms

Table S7 is the originally submitted scan, which used a *single* input realisation; four significant figures were not supportable on that sample and the 0.2% quoted from it came from the five-channel column while the horizon quoted alongside it came from the eight-channel column. Repeating the scan with twelve realisations gives MC (8 ch) $= 2.926 \pm 0.149$, 2.926 ± 0.148, 2.922 ± 0.145 and 2.880 ± 0.116 for the same four intervals. Since the same seeds are used at every T_d, the change is assessed on paired differences: 0.046 ± 0.011 (SEM, $n = 12$, $t = 4.23$), i.e. 1.6% across five decades. That is a real decrease, resolved above both the seed noise and the grid error, and about eight times the originally quoted figure; the qualitative conclusion—the horizon tracks the clock—is unchanged. Table S8 isolates the requirement that actually binds: how much depolarisation of the register between turnovers the memory tolerates.

The linear memory capacity, which is the quantity these claims concern, also converges in the time grid, something the submitted version reported only for the IPC: MC (8 ch) = 2.889, 2.866, 2.861 for n_t = 48, 96, 192 (six seeds), a 0.17% change across the last refinement.

TABLE S8. The criterion: memory against register depolarisation q.

q	T_d/T_{1n}	MC (5 ch)	MC (8 ch)
0.00	0	1.984	2.890
0.30	0.36	1.958	2.667
0.50	0.69	1.927	2.288
0.70	1.20	1.788	2.030
0.90	2.30	1.218	1.977
1.00	∞	1.002	1.002

Memory degrades steadily as the turnover interval approaches the nuclear relaxation time—by $q = 0.5$ a fifth of the eight-channel capacity is already gone, so "essentially intact" overstates it—and collapses to the memoryless floor MC = 1.00 when every turnover starts from a fresh molecule.

Which nuclear relaxation process enforces the criterion. The register propagated between turnovers is a full nuclear density matrix, so memory might in principle require nuclear coherences as well as populations to survive the pause; the binding constant would then be $\min(T_{1n}, T_{2n})$, which at NMR fields is much the more restrictive. Replacing the depolarising channel by a pure-dephasing channel on each nucleus—which destroys nuclear coherence while preserving $\langle I_z \rangle$ exactly—settles it (four seeds):

TABLE S9. The register is a population register, not a coherence register.

q	MC (8 ch), depolarising	MC (8 ch), dephasing
0.0	2.890	2.890
0.3	2.667	2.919
0.5	2.288	2.899
0.7	2.030	2.878
0.9	1.977	2.867
1.0	1.002	2.865

Complete nuclear dephasing every cycle costs less than 1% of the capacity. The memory resides in nuclear z-populations—precisely what the CIDNP readout reports—so the constant that bounds the horizon is T_{1n}.

This is a statement about what the reservoir uses, not a rescue. At the geomagnetic field the register is in extreme narrowing, where $J(0)$, $J(\omega)$ and $J(2\omega)$ coincide and therefore T_{1n} and T_{2n} agree to one part in 10^7 (Sec. S12); the population/coherence distinction changes nothing there, and would matter only in an NMR magnet.

Regularisation and counting statistics. The ridge parameter $\lambda = 10^{-6}$ on unstandardised features is load-bearing, because the time-resolved channels are nested partial integrals of a single signal and the Gram matrix is correspondingly ill-conditioned ($\kappa \approx 10^9$). For the kinetic route the recovered IPC runs 4.53, 4.11, 1.98, 1.01, 1.00 at $\lambda = 10^{-12}$, 10^{-9}, 10^{-6}, 10^{-3}, 10^{-1}; standardising the features gives 4.53, 4.53, 4.34, 2.26, 1.42. The reported capacities are therefore conservative with respect to this choice, not inflated by it. The physically meaningful regulariser is the noise on a real product pool: adding Gaussian shot noise of relative size $N^{-1/2}$ to each channel leaves the CIDNP capacity at 4.54 for $N = 10^{10}$, 4.54 for $N = 10^8$ and 3.93 for $N = 10^6$, so at cellular copy numbers the readout is not shot-noise limited.

S8. A SEMICLASSICAL REFERENCE FOR THE COHERENCE FRACTION

The coherence fraction of Sec. S5 is defined by an operational dephasing limit, not by comparison against a physically motivated classical model, and we said there that it should not be read as a quantum/classical decomposition without one. This section supplies the reference.

We keep the Hamiltonian, couplings, drive, readout and estimator identical and change one thing: every spin becomes a classical vector of length $\sqrt{S(S+1)}$ precessing in the field of the others, $\dot{\boldsymbol{S}}_i = \boldsymbol{S}_i \times \partial H / \partial \boldsymbol{S}_i$. This is the Schulten–Wolynes/Timmel semiclassical treatment carried to its natural conclusion—classical nuclei *with* back-action, so that the register can still carry memory between turnovers. A static-nucleus semiclassical model would have no memory by construction and would be a straw man. The correspondences are exact in the mean: $\langle P_S \rangle = \frac{1}{4} - \boldsymbol{S}_1 \cdot \boldsymbol{S}_2$, singlet birth is an antiparallel pair ($P_S = 1$), triplet birth is a fixed angle $\cos\theta = 1/3$ ($P_S = 0$), and electronic T_2^e becomes a white-noise z-field. Each of $M = 512$ trajectories is one molecule carrying its own register, so the trajectory average is simultaneously the classical reference and a pooled ensemble of independent registers.

Two checks: the sampled $\langle P_S \rangle$ reproduces the input s to four decimals across $s \in [0, 1]$, and $|\boldsymbol{S}|$ is conserved to 5×10^{-7} over a cycle. The capacity shows no systematic drift with the integration step (0.78, 0.88, 0.73 for 3, 6 and 12 substeps per output point); the scatter is Monte-Carlo noise in the pooled observable, not integration error.

At the engineered point with the same twelve observables (four seeds, $L = 700$):

$$\mathrm{IPC}_{\mathrm{classical}} = 0.764 \pm 0.015, \qquad \mathrm{IPC}_{\mathrm{quantum}} = 5.421,$$

a coherence fraction of 0.859 against the 0.816 obtained from the operational dephasing limit. The classical value sits slightly below the nominal memoryless floor of unity because of sampling noise in the $M = 512$ pooled observable, so the comparison that matters is against the classical model's *own* floor, obtained by randomising its register every cycle:

$$\mathrm{IPC}_{\mathrm{classical}} = 0.764 \pm 0.015, \quad \mathrm{IPC}_{\mathrm{classical}}^{q=1} = 0.768 \pm 0.016.$$

The excess is -0.004: zero within the noise. The classical reservoir retains *no* memory of previous turnovers at all, and its linear and total capacities coincide to three decimals, so it has no nonlinear capacity either. Classical nuclei with full back-action, given every opportunity to store information between cycles, store none.

The conclusion is the one the main text hedged against, and it goes the paper's way: measured against a genuine classical model of the same spin system, rather than against an operational dephasing limit, the coherence fraction is if anything *larger* than reported. The 0.82 quoted in the main text is conservative.

S9. WHICH NUCLEI CAN HOLD THE REGISTER

The main text applies a single relaxation probability q to the whole nuclear register, with $T_{1n} = 1$ s. The relaxation literature does not support treating the three nuclei alike, and the correction matters more than any other in this work.

What is measured. For protein-bound ^{1}H the assumption is sound: non-selective amide T_1 in ubiquitin is 1.15 s at 600 MHz [Tjandra *et al.*, J. Am. Chem. Soc. **117**, 12562 (1995)], and 3.0–18.7 s in a deuterated domain [Ulmer *et al.*, J. Magn. Reson. **166**, 190 (2004)]. In vivo ^{31}P in human brain and muscle spans 1.1–6.7 s [Bogner *et al.*, Magn. Reson. Med. **62**, 574 (2009); Ren *et al.*, NMR Biomed. **28**, 1455 (2015)]. For ^{14}N it is not sound. Nitrogen-14 is spin-1 and relaxes quadrupolarly; in immobilised proteins ^{14}N acts as a fast-relaxing relaxation sink [Sunde and Halle, J. Magn. Reson. **203**, 257 (2010)], and rotationally immobilised protein protons already show $1/T_1 > 2000$ s^{-1} at low field [Goddard, Korb and Bryant, J. Magn. Reson. **199**, 68 (2009)]. Using the measured quadrupole tensors of the flavin semiquinone—$e^2qQ/h \approx 3.6$ MHz for N5 and 4.8 MHz for N10 [Martínez *et al.*, Magn. Reson. **6**, 183 (2025)]—with a rotational correlation time of ~ 20 ns for a cryptochrome-sized protein, and noting that at geomagnetic field $\omega\tau_c \ll 1$ places the system firmly in extreme narrowing where quadrupolar relaxation is *maximal*, gives $T_1(^{14}\mathrm{N}) \approx 0.14$–$0.26$ μs. Even the most favourable measured biological ^{14}N coupling reaches only ~ 1.3 μs.

Two of the three nuclei in our model are ^{14}N5 and ^{14}N10 on FAD. For them, $q = 1 - \exp(-T_\mathrm{d}/T_{1n}) \to 1$ for any turnover interval beyond a microsecond: they are wiped every cycle, whatever the turnover rate.

We note for completeness that the objection usually raised—paramagnetic relaxation from dissolved O_2 and cytoplasmic metal ions—is *not* what binds here. Measured oxygen relaxivity is 0.14 s^{-1}mM^{-1} on water protons at ≥ 3 T [Vatnehol *et al.*, MAGMA **33**, 447 (2020)] and 0.03–0.5 s^{-1}mM^{-1} on protein-bound protons depending on burial [Teng and Bryant, Biophys. J. **86**, 1713 (2004)]; at a tissue [O_2] of ~ 30 μM, and with cytosolic labile Fe at 0.2–7 μM, the total paramagnetic contribution is $\lesssim 0.1$ s^{-1}, a ceiling of tens of seconds. The bound on the proton register is set by slow tumbling, not by paramagnets.

What survives. We therefore relax each nucleus independently, depolarising nucleus j by $\mathrm{Tr}_j(\rho) \otimes \mathbb{1}_j/2$, and score every scenario against the all-wiped floor of the same readout (six seeds).

TABLE S10. Capacity by which nuclei retain their polarisation, as excess over the all-wiped floor.

register retained	kinetics	excess	CIDNP	excess
all three (main text)	2.008	+0.989	4.650	+2.644
^{1}H only (literature)	1.019	+0.000	2.784	**+0.779**
^{14}N only	1.020	+0.001	3.447	+1.442
none (floor)	1.019	0	2.005	0

The conclusion is a real weakening of the paper, and we state it as such. A physically defensible register—the proton alone—retains +0.779 of excess capacity, 29% of the value obtained when all three nuclei are assumed to persist. The time-resolved kinetic readout is destroyed outright: it needs the whole register and returns exactly its floor when either subset is wiped. (That is a statement about the survivor register: with the product register the kinetic excess still falls to zero when the ^{14}N are wiped, but wiping the proton alone leaves +0.48.) **The reservoir therefore survives on the proton register, at roughly a third of the capacity, and only through the CIDNP channel.** The proton T_{1n} of ~ 1 s that makes this seem comfortable is, however, a *high-field* measurement; Sec. S12 shows that at the geomagnetic field the same nucleus relaxes some four hundred times faster, which tightens the criterion further. What shrinks here is how much the reservoir can compute and how it must be read; what Sec. S12 shrinks is the window in which it can be read at all.

This also sharpens the experimental target: the criterion should be tested on a radical pair whose register is protons, and the relevant relaxation time is the proton T_1 of the diamagnetic product, not a generic "nuclear T_1".

S10. ENSEMBLE AVERAGING

Every capacity above is a single-molecule quantity, whereas a downstream reaction reads a pool of order 10^{10} copies. A density matrix already averages over a *homogeneous* ensemble sharing a common clock, so that case is covered; what is not covered is heterogeneity between copies and desynchronisation of their turnover schedules. We compute both by pooling the product signal of many copies driven by a common input, and—as with the accumulated-pool route—we measure each pooled readout against its *own* floor, obtained by wiping the nuclear register every cycle. Only the excess over that floor is register memory.

Desynchronisation is the binding constraint. Copies are placed at phase offsets drawn uniformly in $[-\Delta, +\Delta]$ cycles (64 copies, three seeds). A copy offset by more than one cycle is showing an earlier *input*, so pooling itself becomes a delay line; the floor control is essential.

TABLE S11. Pooled capacity against turnover jitter Δ, as excess over the register-wiped floor of the same pooled readout.

Δ (cycles)	kinetics IPC	excess	CIDNP IPC	excess
0	2.000	+0.991	4.660	+2.664
0.1	1.790	+0.781	3.510	+1.901
0.25	1.607	+0.598	3.377	+1.756
0.5	1.015	+0.006	3.140	+1.518
1	1.119	+0.110	3.176	+1.555
2	1.885	+0.004	3.833	+1.310

The two readouts behave completely differently. The time-resolved kinetics lose almost all of their memory by half a cycle of jitter with the survivor register (+0.991 → +0.006) and reach zero near one and a half cycles with the product register (+1.716 → +0.317 at half a cycle, +0.181 at one, +0.016 at one and a half; Sec. S4). Those channels sample the cycle at particular times, so a pool spread over the cycle averages them into a single number. The survivor sequence is not monotonic—it returns to +0.110 at one cycle—and the $q = 1$ floor rises beyond one cycle, so these numbers should be read as the excess they are and not as a smooth decay curve. The apparent recovery of the raw IPC at $\Delta \geq 1$ is an artifact of pooling and not memory at all—at $\Delta = 2$ the excess over the register-wiped floor is +0.004. The CIDNP route, by contrast, retains half its excess even at $\Delta = 2$, because the product-weighted polarisation is integrated over the whole cycle and does not depend on when the pool is inspected. **Synchronised turnover is therefore a requirement of the kinetic readout specifically, not of the reservoir.** Of the two routes evaluated here, CIDNP is the one that survives a desynchronised ensemble; the remaining five routes of Table S2 were not evaluated under jitter, so we do not claim it is the only survivor among all seven.

Heterogeneity is not a threat. Scattering the three hyperfine couplings log-normally about the cryptochrome point does not degrade the pooled capacity; see `open1_ensemble_heterogeneity.json`. Mild spread slightly *improves* it, which we attribute to the averaging reducing the ill-conditioning of the nested time-resolved channels discussed above rather than to any spin-dynamical effect.

Heterogeneous relaxation. The heterogeneity above varies the Hamiltonian at a common relaxation rate, which is not the variable the register criterion turns on. A real pool does not tumble at one rate: some copies stay bound to the protein and some diffuse freely, so τ_c—and through it T_{1n}, and through that the fraction of the register surviving a turnover—differs from copy to copy. Because the readout sums over copies and cannot tell them apart, a fast-relaxing sub-population could in principle wash out a slow one's memory. We test this by drawing τ_c log-normally about a 5 ns median with log-spread σ, taking six stratified quantiles as the copies, and giving each its own survival $q_j = 1 - \exp(-T_\mathrm{d}/T_1(\tau_c^{(j)}))$ through the relaxation model of Sec. S12 at $T_\mathrm{d} = 10$ ms (three seeds, eight-channel CIDNP route).

TABLE S12. Pooling copies that relax at different rates. The mean-field column is a single homogeneous copy at the pool's own mean q; the last two columns are what the *spread* contributes once that is removed, in absolute terms and as a fraction of the rise over $\sigma = 0$.

σ	τ_c (ns)	$\bar{q}$	pooled excess	spread alone	of the rise
0	5.00	0.739	+1.956	−0.000	—
0.3	3.30–7.57	0.735	+1.962	+0.003	46%
0.6	2.18–11.46	0.722	+1.985	+0.017	57%
1.0	1.25–19.93	0.696	+2.044	+0.059	66%

The pooled excess does not fall; it rises slightly, from +1.956 to +2.044 across a sixteen-fold spread in τ_c—a change of 4.5 per cent. Two parts of the rise can be separated. The mean survival $\bar{q}$ drifts from 0.739 to 0.696, and a lower q leaves more register intact; the mean-field control isolates that part. The drift is not because a wider distribution holds more slow copies—the copies are stratified quantiles, so three lie on each side of the median at every σ—but because $q = 1 - e^{-T_\mathrm{d}/T_1}$ saturates at unity. A copy displaced to slow τ_c can lose much more than the fast copy opposite it can gain, so a symmetric spread in τ_c produces an asymmetric, downward response in q: at $\sigma = 1$ the six values are 0.286 to 0.995, i.e. -0.45 and $+0.26$ about the median. At $\sigma = 1$ it accounts for $+0.030$ of the $+0.088$ rise. The remaining $+0.059$ is the spread itself, so the *majority* of the rise—66 per cent—is not a mean-field effect. The same split gives 46 and 57 per cent at $\sigma = 0.3$ and 0.6. At $\sigma = 0$ the control reproduces the pooled value to 8×10^{-13}, which is the consistency check that the decomposition is doing what it claims; the residual is positive at all three non-zero spreads.

What this does and does not license is worth stating carefully. It licenses applying the criterion to a heterogeneous pool: capacity never falls below the uniform value, and the total effect of a sixteen-fold spread stays under five per cent, so a single effective T_{1n} is an adequate working approximation. It does *not* license the cleaner claim that a heterogeneous pool simply is a homogeneous pool at its mean survival—the mean-field control was run precisely to test that, and it fails: most of the (small, favourable) difference is the spread itself, not the shifted mean. The scan covers $\sigma \le 1$ about a single median and does not speak to a pool split into two widely separated populations. It does not rescue the kinetic routes from desynchronisation, which is a separate axis and remains the binding ensemble constraint (`open1_ensemble_desync_floor.json`); the scan of this section is `open6_tau_c_heterogeneity.json`.

S11. ANISOTROPIC HYPERFINE COUPLING AND THE NUCLEAR SPIN QUANTUM NUMBER

The main text uses isotropic couplings and models ^{14}N as spin-1/2. We relax each in turn. Both scans use the same cycle, readout and estimator, and each is scored against its own $q = 1$ floor (three seeds).

Anisotropy. We hold the isotropic average fixed and grow an axial anisotropy, $\mathbf{A} = A_\mathrm{iso}(1-\eta, 1-\eta, 1+2\eta)$, at fixed molecular orientation. This is a *scan*, not a calculation with the published tensor sets [Lee *et al.*, J. R. Soc. Interface **11**, 20131063 (2014); Schleicher *et al.*, Sci. Rep. **11**, 18395 (2021)], and it bounds the sensitivity without removing the limitation.

TABLE S13. Capacity against axial anisotropy η (eight-channel CIDNP route).

η	IPC	floor	excess
0 (isotropic)	4.660	1.996	**2.664**
0.5	2.863	2.001	0.862
1.0	1.009	1.009	0.000
2.0	2.730	1.999	0.731

This is the least comfortable result in the paper. Realistic axial anisotropy costs roughly two-thirds of the capacity, and at $\eta = 1$ it vanishes exactly. The mechanism is transparent and is the same one that makes CIDNP the write channel: $\eta = 1$ is the pure-Ising point $A_{xx} = A_{yy} = 0$, where the hyperfine term commutes with I_z and S_z, so there are no flip-flops, no singlet–triplet mixing driven by the nuclei, and no polarisation transfer. The capacity is carried by the *perpendicular* components of the hyperfine tensor, and $\eta = 2$—where $|A_\perp|$ is restored with opposite sign—recovers to 0.731. A real flavin or tryptophan coupling has substantial $A_\perp$, so the physical conclusion is that the reservoir survives anisotropy with a substantially reduced capacity, and fails only in an Ising limit that no real radical realises. We report the reduction rather than quoting the isotropic number as though it were general.

^{14}N as spin-1. A four-spin reduction (e_1, e_2, one N, one H) is run with the nitrogen as spin-1/2 and then as spin-1, everything else fixed, so the difference isolates the spin quantum number.

TABLE S14. The spin-1/2 approximation for ^{14}N is conservative.

model	dim	IPC	floor	excess
N as spin-1/2 (main text)	16	3.267	1.984	1.283
N as spin-1 (real ^{14}N)	24	3.639	1.997	**1.642**

Treating ^{14}N correctly *increases* the excess capacity by 28%: the larger nuclear Hilbert space is a larger register. The approximation used in the main text therefore understates the reservoir, and no conclusion depends on it.

S12. PREDICTING THE TWO UNMEASURED CONSTANTS

The criterion is stated in terms of $T_\mathrm{turnover}/T_{1n}$ and neither quantity has been measured for a neuronal cryptochrome. Neither can be measured here, but both can be *predicted* from measured constants, and doing so changes the conclusion. We validate the relaxation calculator against three independent measured systems before applying it to the unmeasured one.

TABLE S15. Validation of the relaxation calculator.

system	predicted	measured	ratio
^{2}H in D_2O (quadrupolar)	0.432 s	0.40 s	1.08
^{14}N of free amino acid	2.3 ms	2.9–8.8 ms	0.5
$^{1}H_2O$ intramolecular (dipolar)	5.55 s	3.6 s[1]	1.54

The relaxation time is a low-field quantity. For a 60 kDa protein at 310 K, Stokes–Einstein–Debye gives $\tau_c = 15.2$ ns. At the geomagnetic field $\omega\tau_c \approx 2\times10^{-4}$, so every nucleus here sits deep in extreme narrowing, where $1/T_1 \propto \tau_c$ is *maximal.* This is the point the $T_{1n} \approx 1$ s assumption misses: that value is a high-field number.

TABLE S16. Predicted T_1 of candidate register nuclei.

nucleus	at 50 μT	at 9.4 T
Trp Hβ (CH_2)	2.4 ms	9.0 s
flavin 8α-CH_3	5.0 ms	2.3 s
^{14}N5 / N10 (flavin)	0.34 / 0.19 μs	7 / 4 μs

The protein-bound proton relaxes in milliseconds at the field the organism actually experiences, not in seconds. Because $1/T_1 \propto \tau_c$ in this regime, the register's lifetime is governed by how fast the product reorients: T_1 rises to 37 ms at $\tau_c = 1$ ns and to 0.37 s at $\tau_c = 0.1$ ns.

The drive cannot be light. With the measured flavin extinction coefficient ($\varepsilon_{450} \approx 11\,300$ M^{-1}cm^{-1}, cross-section 4.3×10^{-17} cm^2) and a quantum yield of 0.1, a single flavin turns over once per 7.7 s in full sunlight, once per 77 s under overcast sky, and—for a blue-light attenuation of 10^{-9} through a centimetre of tissue—once per ~ 250 years inside the skull. A light-driven cryptochrome reservoir in a neuron fails by roughly ten orders of magnitude. The drive must therefore be catalytic, and a flavoenzyme turnover number of 1–1000 s^{-1} puts $T_{\rm turnover}$ at 1 ms–1 s, which is the right range.

Putting the two together. Requiring simultaneously that (a) the register be reused, (b) T_1 follow from the calculation above, and (c) the horizon $(\mathrm{MC} - C_0)T_{\rm turnover}$ land in the 10 ms–1 s band, and interpolating MC(q) from the measured reuse curve of Table S8:

TABLE S17. Feasible region. The retained memory is required to exceed 0.5, about four times the seed dispersion, so that the residual $\mathrm{MC}(q{=}1) - C_0 = 0.002$ cannot manufacture a horizon.

τ_c	T_1 (50 μT)	viable $T_{\rm turn}$	horizon
0.1 ns	372 ms	5.4–1340 ms	937 ms
1 ns	37 ms	5.6–134 ms	94 ms
5 ns	7.4 ms	9.8–27 ms	19 ms
10 ns	3.7 ms	none	9.4 ms
15.2 ns (bound)	2.4 ms	**none**	**6.2 ms**

Two corrections are needed before this table can be read as a boundary, and they pull in opposite directions.

The boundary is solved, not read off the grid. The scan above samples τ_c at $\ldots, 5, 10, 15.2$ ns and never evaluates anything between 5 and 10, so quoting "feasible only for $\tau_c \le 5$ ns" reports the last sampled point rather than the boundary. Bisecting on the same feasibility predicate puts it at 9.37 ns, a requirement of 1.6× rather than 3×.

The relaxation calculation omitted the intermolecular bath. Our T_1 counted only the intramolecular partner. The validation table above supplies the scale of that omission: liquid water's intramolecular-only prediction is 5.55 s against a measured total of 3.6 s, a bath contributing 54% of the intramolecular rate. We use that residual as a calibration, not as an independent measurement—f is defined from it, so the ratio it restores to unity is an identity. Its transplant onto a protein interior is defensible by magnitude rather than by mechanism: the resulting $\Sigma_{\rm ext} = 0.017$ Å^{-6} sits at the low end of the 0.016–0.030 a uniform protein proton density gives for closest approaches of 2.5–2.0 Å, which is the conservative end for a claim of exclusion. Restoring it shortens T_1 for the protein-bound proton from 2.4 to 1.6 ms, the ceiling from 6.1 to 4.0 ms, and moves the boundary back from 9.37 to 6.08 ns—a requirement of 2.5×. The two errors very nearly cancel, which is why the published $\tau_c \lesssim 5$ ns and "about three times" are close but not exact—the converged values are $\lesssim 6$ ns and 2.5×—and they were reached by a route that does not support them, and the corrected route is what we now give.

A register whose dipolar vector reorients only with the intact protein cannot reach the neural band: its ceiling is 6.1 ms on the intramolecular term alone and 4.0 ms with the bath, below the band under every turnover rate. The criterion is a statement about $\tau_{\rm eff} = S^2\tau_c + (1-S^2)\tau_{\rm int}$ and about the total dipolar sum, not about chemical binding. The flavin 8α-CH_3 makes the distinction concrete: its intramethyl H–H vector has the exact geometric value $S^2 = 1/4$ and therefore does not co-rotate with the protein despite being covalently attached, which on the intramolecular term alone would give $T_1 = 5.0$ ms and a ceiling of 12.7 ms—inside the band. Its own boundary sits at 19.3 ns without the bath, i.e. it would be feasible even at the protein's own tumbling rate. With the bath that boundary falls to 10.8 ns and the escape closes. The two candidate nuclei do not land at the same requirement—the methyl's boundary is 10.8 ns against 6.1 ns for Trp Hβ, because its intramethyl vector is already partly averaged—but both fall below the protein's own 15.2 ns, the methyl by a factor of 1.4.

How much of this depends on the band edge. The requirement scales linearly with where the neural band is taken to start. In extreme narrowing $1/T_1 \propto \tau_{\rm eff}$, so the ceiling is exactly inversely proportional to $\tau_{\rm eff}$ and the required speed-up is

$$\text{speed-up} = 1.63\,(1+f) \times \frac{\text{band edge}}{10\text{ ms}},$$

with f the intermolecular bath ratio. Bisecting numerically agrees with the closed form to two decimals. A register that reorients only with the protein becomes feasible once the band edge falls below 6.1 ms on the intramolecular term alone, but below only 4.0 ms once the bath is included. Taking the band to start at 5 ms rather than 10 ms therefore does *not* rescue it. We state this because the 10 ms edge is a convention, and the conclusion should be read as conditional on it.

TABLE S18. Robustness of the ceiling for the Trp Hβ vector ($S^2 = 1$). Each listed variation leaves the horizon below the 10 ms band edge; several were already computed for the paper and not reported. The table varies the tumbling time, the partner count and the memory threshold; it does not vary the bath ratio f or the bath order parameter, whose sensitivity is given in the text, nor S^2 itself, which is the separate methyl case above.

variation	τ_c (ns)	T_1 (ms)	ceiling (ms)
as published, hydration 1.3	15.24	2.44	6.15
hydration 1.0	11.73	3.17	7.99
hydration 1.6	18.76	1.98	4.99
viscosity ×2	30.49	1.22	3.07
viscosity ×4	60.97	0.61	1.54
120 kDa dimer	30.49	1.22	3.07
2 dipolar partners	15.24	1.22	3.07
3 dipolar partners	15.24	0.81	2.05
memory threshold 0.2	15.24	2.44	6.15
memory threshold 1.0	15.24	2.44	4.14
+ bath ($f = 0.54$)	15.24	1.58	3.99

Several of these shorten the ceiling further. The one that would *lengthen* it, and so weaken the exclusion, is a rank error: $\tau_{\rm rot}$ here is the *rank-2* correlation time, $1/(6D_r)$, which is what the dipolar spectral densities require. The rank-1 value used in dielectric and fluorescence-anisotropy work is three times longer (45.7 ns), and feeding it into $J(\omega)$ would give $T_1 = 0.81$ ms and a ceiling of 1.8 ms. That mistake would strengthen the exclusion, so we name the rank in the function itself rather than rely on a reader catching it.

This converts the paper's weakest statement—a criterion whose two parameters were simply unknown—into a quantitative and falsifiable one:

The radical pair must be regenerated catalytically, not photochemically, on a register that reorients with $\tau_c \lesssim 6$ ns while remaining part of the same molecule, at a turnover interval drawn from the corresponding row of Table S17: 10–22 ms at $\tau_c = 5$ ns, 6–110 ms at 1 ns, 5–1100 ms at 0.1 ns. The two conditions are not independent, and quoting a single interval alongside a single τ_c bound would overstate the region.

Both halves are measurable with existing methods: τ_c by NMR relaxation dispersion on the isolated protein, and the turnover number by steady-state kinetics. Neither requires a neuron.

S13. SPIN PARAMETERS

Each radical carries its own nuclei; they are not shared, which is what a spatially separated pair means. The Hamiltonian therefore contains three hyperfine terms, two on the first radical and one on the second. An earlier version of the code accepted a fourth hyperfine argument that was never added to the Hamiltonian and had no effect on any result; it has been removed rather than left in place.

TABLE S19. Hyperfine assignment at the cryptochrome point. Values are representative isotropic magnitudes of the dominant $FAD^{\bullet-}$ and $TrpH^{\bullet+}$ couplings; the full anisotropic tensors are given by Lee *et al.*, J. R. Soc. Interface **11**, 20131063 (2014), and the flavin-radical ENDOR couplings by Schleicher *et al.*, Sci. Rep. **11**, 18395 (2021).

coupling	radical / nucleus	(mT)	(MHz)
A_{e_1a}	$FAD^{\bullet-}$, ^{14}N5	~ 0.50	14
A_{e_1b}	$FAD^{\bullet-}$, ^{14}N10	~ 0.18	5
A_{e_2a}	$TrpH^{\bullet+}$, Hβ1	~ 1.43	40

The engineered operating point uses $A_{e_1a} = 80$, $A_{e_1b} = 40$, $A_{e_2a} = 15$ MHz, $J = 2$ MHz, $B = 1$ mT and $\tau = T_2^e = 50$ ns. The conclusions depend on these magnitudes lying in the few–40 MHz range typical of flavin and tryptophan radicals, not on the precise values. The register criterion is *not* independent of the hyperfine assignment, however: Sec. S11 shows that the capacity is carried by the perpendicular components of the hyperfine tensor and vanishes when they do.